

Points on the Line

E L A + M A T H · W O R D S T U D Y

Mr. H Math · Day 24

WRITE YOUR NAME — one letter per box, straight and neat

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Date: _____

Score: ____ / 14

Maya's smoothie uses **2 cups of strawberries for every 3 cups of yogurt**. Every row of her ratio table is an **ordered pair**. Plot them and they line up on one straight line through **(0, 0)**. A ratio made by **adding** falls **off** the line.

DO NOW · SILENT START

3 min

Which table doesn't belong? Circle it and explain.

A	2:3	4:6	6:9
B	1:5	2:10	5:25
C	3:4	5:6	7:8

Because _____

TURN & TALK

2 min

If Maya uses 0 cups of strawberries, how much yogurt does she use? Where is that point on a graph?

Frame: "Zero ____ means zero ____, so the point is (__, __)."

SECTION A · WORD BANK

ordered pair origin x-coordinate y-coordinate equivalent

- (4, 6) is an _____. Move across first, then up.
- The point (0, 0) is called the _____.
- In (4, 6), 4 is the _____ and 6 is the _____.
- Ratios whose points are on the same line through (0, 0) are _____.

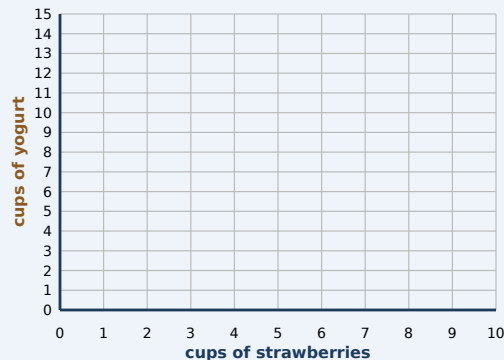
SECTION B · TRY IT: TABLE → POINTS → LINE

Maya's smoothies — complete the table, then plot every point and draw the line from (0, 0).

Batches	1	2	3	4
Strawberries (x)	2			
Yogurt (y)	3			
Point	(2, 3)			

Every step goes right ____, up ____.

Use the line: how much yogurt goes with **5 cups** of strawberries?



SECTION C · ON THE LINE OR OFF THE LINE?

The recipe is **3 cups flour for every 2 eggs**. Decide, then prove it with numbers (multiply or cross products).

Point	On / Off	Proof
(6, 4)		
(5, 4)		
(12, 8)		
(7½, 5)		

Jordan's error: "3 : 2 = 5 : 4 because I added 2 to both." Explain in one sentence why (5, 4) is off the line.

REFINE · SUMMARY — Every row of a ratio table is an ordered pair (x, y). Equivalent ratios sit on **one straight line through (0, 0)**. Multiply to stay on the line; adding moves you off it. Use the line to read missing values between the table rows.

★ Find the multiplier, keep the order, check it a second way! ★